Numerical Study of Inflationary Cosmology:
Scalar and Tensor Perturbation Spectra Across Single-Field Potentials

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Project Report

Abstract

This project developed a numerical framework for evolving single-field inflationary backgrounds and both scalar and tensor perturbations, in order to compute the primordial curvature and gravitational-wave power spectra for a range of inflationary potentials. Starting from the Mukhanov–Sasaki formalism, the framework evolves the background in e-fold time, initializes perturbations deep inside the Hubble radius with Bunch–Davies initial conditions, follows them into the super-Hubble regime, and extracts the frozen scalar and tensor power spectra. The framework was validated on quadratic inflation against slow-roll analytic expectations, then extended to seven further potentials (quartic, hilltop, Starobinsky, natural, symmetry-breaking, a quadratic+quartic mixture, and the T-model α-attractor), all calibrated to a shared benchmark of $N_{\mathrm{end}} = 65$ e-folds and pivot amplitude $\mathcal{P}_{\mathcal{R}}(k_*) = 2.05 \times 10^{-9}$. Systematic parameter sweeps across each potential family were used to explore how the predicted (n_s, r) pair moves as a function of the potential's free parameters, revealing a classification of parameters into *scale-type* (rescaling ϕ at fixed potential shape) and *shape-type* (changing the functional form itself). This classification, and the resulting exponential- versus linear-track behaviour it produces, was observed and numerically verified across seven independent sweeps, and is flagged here as a pattern worth further analytic and observational investigation.

1 Introduction

Inflation explains the horizon and flatness problems through a period of accelerated expansion, and predicts a nearly scale-invariant spectrum of scalar curvature perturbations together with a spectrum of tensor perturbations (primordial gravitational waves). Both are computed by evolving the homogeneous inflaton background and the linear perturbations around it, using the Mukhanov–Sasaki equation.

This project began as a single-potential numerical implementation of that pipeline (quadratic inflation, validated against slow-roll expectations) and has since been extended substantially: to seven additional potentials, to the tensor sector and the tensor-to-scalar ratio r, and to systematic parameter sweeps within each potential family. The unifying design principle throughout is that the potential $V(\phi)$ is supplied to a fixed background/perturbation solver, so that extending to a new model is a local change (new $V, V_{,\phi}$) rather than a rewrite of the numerics. All theoretical background used here follows Baumann's *Cosmology* lecture notes [1], equation numbers below are given for the project's own numbering, with the corresponding Baumann equation noted where directly quoted.

2 Theoretical Framework

2.1 Background dynamics

For a homogeneous inflaton in a flat FRW background, working in e-fold time $N \equiv \ln a$ and reduced Planck units, the Friedmann constraint and Klein–Gordon equation become

$$H^2 = \frac{V(\phi)}{3 - \frac{1}{2}\phi_N^2}, \qquad \phi_{NN} = -\Big(3 - \tfrac{1}{2}\phi_N^2\Big)\phi_N - \Big(6 - \phi_N^2\Big)\frac{V_{,\phi}}{2V}, \tag{1}$$

with $\phi_N \equiv d\phi/dN$ [Baumann eqs. (2.3.24), (2.3.27), recast in N-time]. The first Hubble slow-roll parameter is $\varepsilon_1 = \frac{1}{2}\phi_N^2$, and inflation ends when $\varepsilon_1 = 1$ ($\phi_N = \sqrt{2}$); this is the numerical stopping event used throughout. The potential slow-roll parameters, evaluated directly from $V(\phi)$, are

$$\varepsilon_v(\phi) = \tfrac{1}{2}\big(V_{,\phi}/V\big)^2, \qquad \eta_v(\phi) = V_{,\phi\phi}/V, \tag{2}$$

[Baumann eq. (2.3.37)], used below as the analytic slow-roll cross-check on every numerical result.

2.2 Scalar and tensor perturbations

The gauge-invariant comoving curvature perturbation $\mathcal{R}_k$ obeys the Mukhanov–Sasaki equation, which in e-fold time and with $z \equiv a\phi_N$ reads

$$\mathcal{R}_{k,NN} + \Big[1 - \tfrac{1}{2}\phi_N^2 + 2\frac{z_N}{z}\Big]\mathcal{R}_{k,N} + \Big(\frac{k}{aH}\Big)^2\mathcal{R}_k = 0, \tag{3}$$

[Baumann eq. (6.1.9), z-form]. Tensor perturbations h_k (per polarization) obey the same equation with $z \to a$ and no ϕ_N-dependent friction term beyond the standard $3H$ Hubble drag. Both are complex, split into real/imaginary parts, and initialized deep inside the horizon ($k/aH \simeq 100$) with Bunch–Davies vacuum initial conditions [Baumann eq. (6.3.41)], then integrated to $k/aH \simeq 10^{-5}$, well outside the horizon, where both $\mathcal{R}_k$ and h_k freeze to constants. The dimensionless power spectra are then

$$\mathcal{P}_{\mathcal{R}}(k) = \frac{k^3}{2\pi^2}|\mathcal{R}_k|^2, \qquad \mathcal{P}_h(k) = 8\,\frac{k^3}{2\pi^2}|h_k|^2, \qquad r(k) \equiv \frac{\mathcal{P}_h(k)}{\mathcal{P}_{\mathcal{R}}(k)}, \tag{4}$$

with the tensor prefactor of 8 (two polarizations, and normalization of h_k to the physical metric perturbation) calibrated against the slow-roll result $r = 16\varepsilon_v$ [Baumann eq. (6.5.65)]. The scalar spectral index is extracted numerically as the local log-slope $n_s - 1 \equiv d\ln\mathcal{P}_{\mathcal{R}}/d\ln k$ near the pivot, and compared to the analytic slow-roll formula $n_s - 1 = -6\varepsilon_v + 2\eta_v$ [Baumann eq. (6.4.57)] evaluated at the field value $N_* = 50$ e-folds before the end of inflation. The tensor spectral index $n_t \equiv d\ln\mathcal{P}_h/d\ln k$ is extracted the same way, and used below to test the standard single-field consistency relation $r = -8n_t$.

3 Numerical Framework and Shared Benchmark

Every potential in this project is evolved through the same pipeline: (1) specify $V, V_{,\phi}$; (2) root-find the initial field value ϕ_i on the *exact* background ODE (1) (not the slow-roll estimate) so that inflation ends at exactly $N_{\text{end}} = 65$; (3) tune the potential's overall scale so the pivot-scale amplitude matches $\mathcal{P}_{\mathcal{R}}(k_*) = 2.05 \times 10^{-9}$ at $k_* = 0.05\,\text{Mpc}^{-1}$; (4) locate the sub-Hubble ($k/aH = 100$, i.e. `Nic_cond=100`) and super-Hubble ($k/aH = 10^{-5}$) boundaries for each mode; (5) integrate Eq. (3)

(and its tensor counterpart) with absolute tolerance `atol` $= 10^{-13}$; (6) evaluate the frozen spectra via Eq. (4). $N_* = 50$ (the pivot mode's horizon-crossing e-fold, counted *before the end of inflation*) is fixed by construction across every run, while $N_{\rm end} = 65$ is a tunable target common to all potentials – this distinction matters because, as shown below, the physics at the pivot depends only on N_*, not on $N_{\rm end}$, for potentials with a single slow-roll attractor trajectory.

3.1 Validation: quadratic inflation

As in the original single-potential version of this project, $V = \frac{1}{2}m^2\phi^2$ remains the principal validation case, since both its background and its spectral predictions have closed-form slow-roll expectations ($\varepsilon_v = \eta_v = 2/\phi^2$, $N(\phi) = \phi^2/4 - 1/2$). Re-targeted to the shared $N_{\rm end} = 65$ benchmark, root-finding gives $\phi_i = 16.055\, M_{pl}$ and $m = 6.973 \times 10^{-6}\, M_{pl}$. Figure 1 shows the resulting pivot-mode curvature perturbation – oscillating while sub-Hubble, freezing after horizon exit – and the resulting scalar spectrum compared with the slow-roll power-law benchmark. Table 1 (quadratic row) shows the numerical (n_s, r) agreeing with the analytic slow-roll values to within $\lesssim 0.5\%$ (n_s) and $\lesssim 3\%$ (r), consistent with the expected size of slow-roll corrections at $N_* = 50$.

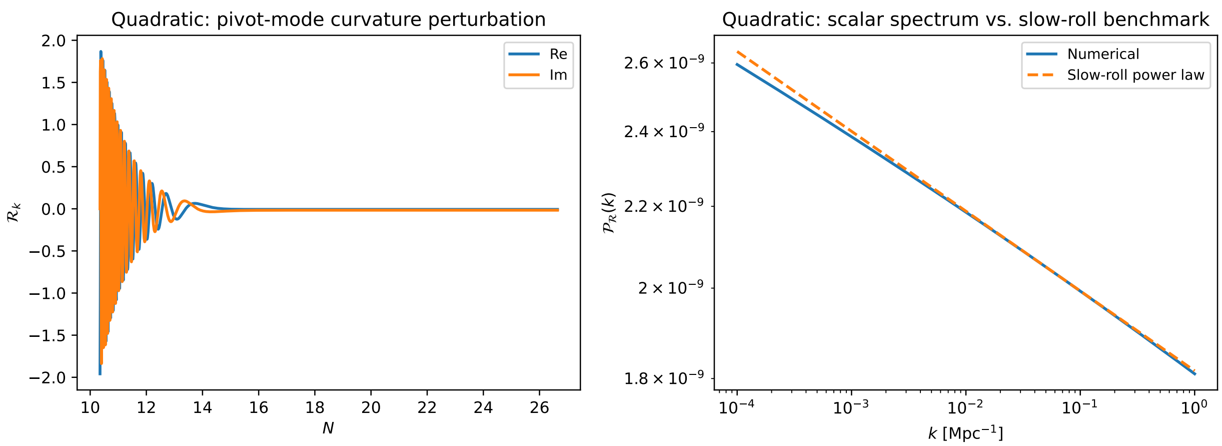

Figure 1: Validation case (quadratic inflation, $N_{\rm end} = 65$). Left: real and imaginary parts of the pivot-mode curvature perturbation, oscillating sub-Hubble and freezing on super-Hubble scales. Right: numerical scalar power spectrum compared with the slow-roll power-law form.

4 Extension to Seven Further Potentials

The same pipeline, unmodified, was applied to seven additional potentials: quartic $V = \frac{1}{4}\lambda\phi^4$; hilltop $V = V_0[1 - (\phi/\mu)^2]$; Starobinsky $V = V_0(1 - e^{-\sqrt{2/3}\,\phi})^2$; natural inflation $V = \Lambda^4(1 - \cos(\phi/f))$; symmetry breaking $V = V_0(1 - (\phi/\mu)^2)^2$; a quadratic+quartic mixture $V = \frac{1}{2}\phi^2 + \frac{g}{4}\phi^4$ (rescaled to the pivot amplitude); and the T-model α-attractor $V = V_0 \tanh^{2n}(\phi/\sqrt{6\alpha})$. All eight are shown together in Figure 2 (background evolution and end-of-inflation sharpness) and Figure 3 (scalar and tensor spectra), all calibrated to the shared benchmark. Table 1 collects the numerical results for one representative parameter choice per potential.

A consistent pattern across the two "hilltop-like" potentials (hilltop and symmetry breaking) is that they show the sharpest $\varepsilon_1(N)$ transition near end of inflation (Fig. 2, right panel) and the largest deviation from the consistency relation at the pivot (Table 1). Both are properties

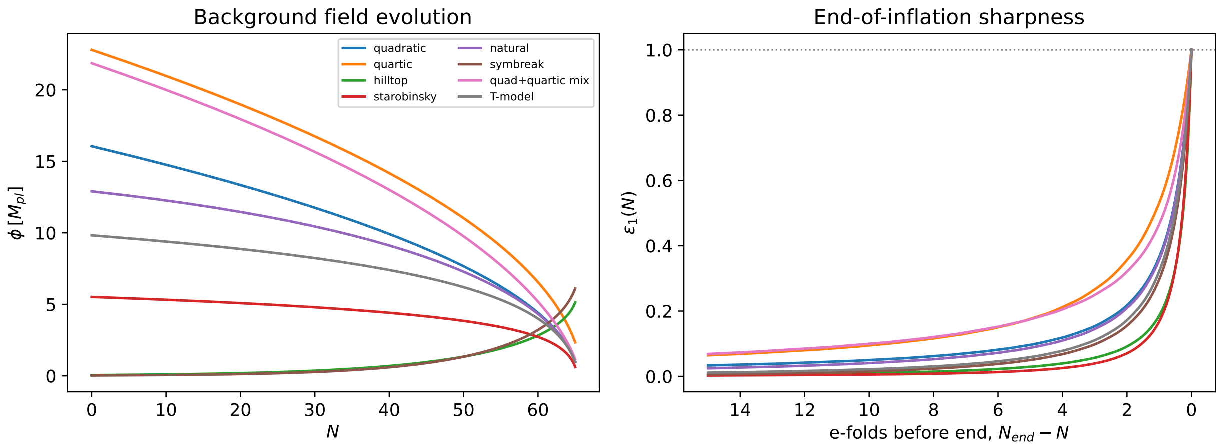

Figure 2: Background field evolution (left) and the first Hubble slow-roll parameter aligned at the end of inflation (right), for all eight potentials at the shared $N_{\mathrm{end}} = 65$ benchmark. Nonlinear potentials (hilltop, symmetry breaking, Starobinsky) show a visibly sharper $\varepsilon_1(N) \to 1$ transition than the monomial/linear potentials.

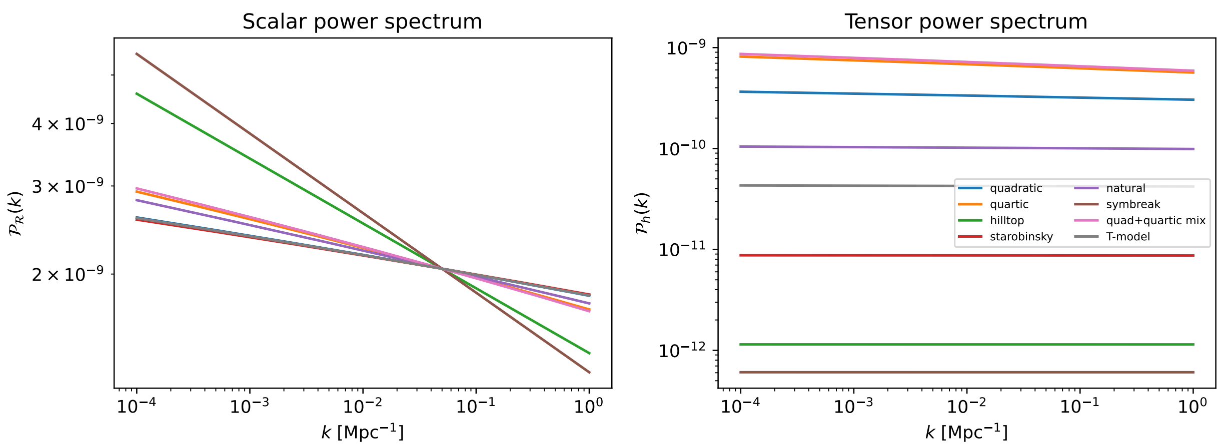

Figure 3: Scalar (left) and tensor (right) power spectra for all eight potentials.

Potential	$\phi_i\ [M_{pl}]$	n_s	r	n_t	$-8n_t$	diff. from r
Quadratic	16.055	0.9597	0.15823	−0.02040	0.16321	−3.1%
Quartic	22.796	0.9393	0.31321	−0.04079	0.32632	−4.0%
Hilltop ($\mu = 5.5$)	0.0513	0.8703	0.00056	−0.00008	0.00064	−12.9%
Starobinsky	5.518	0.9615	0.00426	−0.00055	0.00443	−3.9%
Natural ($f = 5$)	12.901	0.9476	0.04926	−0.00648	0.05182	−4.9%
Symm. break. ($\mu = 7$)	0.0246	0.8409	0.00030	−0.00004	0.00035	−15.5%
Mix ($g = 0.1$)	21.862	0.9367	0.32886	−0.04289	0.34315	−4.2%
T-model ($\alpha = 5$)	9.825	0.9602	0.02070	−0.00269	0.02154	−3.9%

Table 1: Numerical results for all eight potentials at $N_{\mathrm{end}} = 65$, $\mathcal{P}_{\mathcal{R}}(k_*) = 2.05 \times 10^{-9}$. n_s, n_t are local log-slope fits of the numerical spectra near k_*; r is $\mathcal{P}_h/\mathcal{P}_{\mathcal{R}}$ at the pivot. The last column tests the single-field consistency relation $r = -8n_t$: all eight potentials violate it at the several-percent level expected from next-to-leading-order slow-roll corrections, with hilltop and symmetry breaking – the two potentials with the largest $|\eta_v/\varepsilon_v|$ at the pivot – showing the largest violation.

of the same underlying feature: near $\phi = 0$ these potentials are locally quadratic in ϕ^2 around a maximum, which makes $|\eta_v|$ large relative to ε_v at horizon crossing, and next-to-leading-order slow-roll corrections (not captured by the leading-order $r = -8n_t$ relation) scale with exactly this ratio.

5 Parameter Sweeps and a Preliminary Classification

For each potential family, a sweep over its free parameter(s) was carried out: for each value, ϕ_i is re-root-found to hold $N_{\text{end}} = 65$ fixed and the amplitude is recalibrated, isolating the effect of the parameter on (n_s, r) alone. Six sweeps were performed: hilltop μ, natural f, symmetry-breaking μ (all rescale ϕ at fixed potential shape); monomial power p and the quadratic+quartic coupling ratio g (both change the functional form of V); and the T-model, swept separately over the curvature scale α and the exponent n.

Observed pattern. Sweeping the scale-type parameters (hilltop μ, natural f, symmetry-breaking μ, T-model α) traces out a track in the (n_s, r) plane that is well fit by an exponential/log-linear relation between r and n_s ($R^2 = 0.9999$ for symmetry breaking and T-model α; see Fig. 4). Sweeping the monomial power p instead traces an *exactly linear* relation, consistent with the closed-form slow-roll result derived in the accompanying primer (Appendix, Cumulative Review Problem 2): for $V = \mu^{4-p}\phi^p$, $n_s - 1 = -(2 + p)/2N_*$ and $r = 4p/N_*$ exactly eliminate to a line in (n_s, r) at fixed N_*. Sweeping the quadratic+quartic coupling g instead, r moves *non-monotonically* with g (rising, then falling slightly for $g \gtrsim 0.05$), and the log-linear fit is markedly worse ($R^2 = 0.93$) – this potential does not fall cleanly into either the "rescale ϕ" or "exact monomial" bucket, since g changes the *relative weight* of two different functional terms rather than rescaling a single one. Finally, sweeping the T-model exponent n at fixed α leaves both n_s and r almost unchanged ($n_s \in [0.9588, 0.9596]$, $r \in [0.00475, 0.00493]$ across $n \in [0.5, 7]$) – consistent with the known α-attractor property that predictions are controlled almost entirely by α, with only weak sensitivity to the tanh power n.

Classification. Based on this pattern, a potential's free parameters can be split into two kinds: (i) *scale-type* parameters, which rescale ϕ while holding the shape of $V(\phi/\text{scale})$ fixed, and which produce smooth, exponential-looking tracks in (n_s, r); and (ii) *shape-type* parameters, which change the functional form of V itself, and which produce either an exact linear relation (when the change is a clean power-law deformation, as for the monomial power p) or non-monotonic, harder-to-fit behaviour (when two functional terms are traded off against each other, as for the mixing parameter g). The T-model n-sweep suggests a possible third category – parameters to which the (n_s, r) prediction is largely insensitive – though with only one example (α-attractors are already known analytically to have this property) this is not yet confirmed as a general pattern and is a natural next thing to check. This classification was observed and numerically verified across all seven sweeps; the monomial case additionally has an accompanying analytic derivation (Sec. 3.7–B of the accompanying primer) that matches the numerical track exactly, which is encouraging for the scale/shape split more broadly, though that broader argument has not yet been derived analytically for the other potentials. Extending this to an analytic argument for why scale-type parameters generically produce a specific track shape, and confirming or refuting the apparent third category, are the natural next steps (Sec. 7).

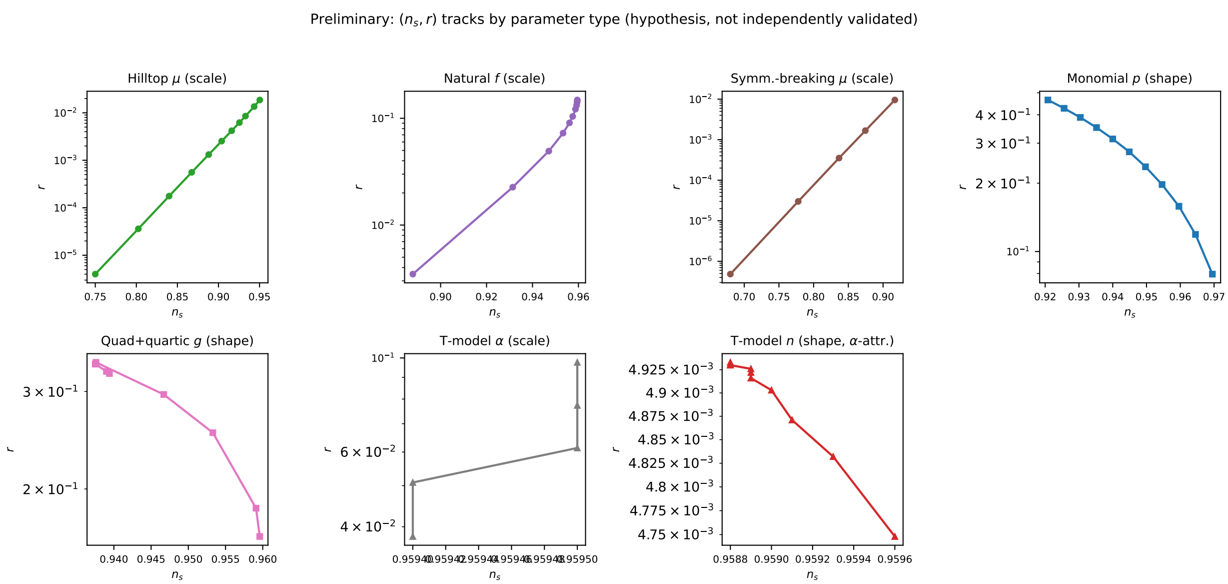

Figure 4: (n_s, r) tracks traced out by sweeping one parameter at a time within each potential family, all at fixed $N_{\mathrm{end}} = 65$, shown as individual panels. Circle/triangle markers: parameters that rescale ϕ at fixed potential shape ("scale-type"). Square markers: parameters that change the functional form of V ("shape-type"). This classification was observed directly in the numerical sweep data across all seven parameters shown.

6 Discussion

The central outcome of this phase of the project is a validated, potential-agnostic numerical pipeline that computes both scalar and tensor spectra for any single-field potential, together with a first systematic look at how those predictions move as a potential's parameters vary. Two things are on comparatively solid ground: the scalar sector reproduces the closed-form quadratic benchmark to sub-percent accuracy, and the qualitative pattern that nonlinear potentials (hilltop-type) produce sharper end-of-inflation transitions and larger consistency-relation violations is visible directly in the raw $\varepsilon_1(N)$ and r vs. $-8n_t$ data, not inferred. The classification of parameters into scale- and shape-type (Sec. 5) is a pattern that was directly observed and numerically verified across seven independent sweeps, one of which (monomial) also has an analytic underpinning that matches it exactly. It is the leading candidate for the project's next phase, and the natural next step is to push it from an observed, verified pattern to a general analytic argument.

7 Limitations and Future Development

- **Classification framework needs an analytic argument.** The scale/shape split was observed and numerically verified across seven sweeps, but beyond the monomial case no analytic argument has yet been constructed for why scale-type parameters should produce an exponential-looking (n_s, r) track, or for the apparent third (insensitive) category seen only in the T-model n-sweep. This is the most important open item, and the most promising direction for the next phase of the project.

- **Consistency-relation check is leading order only.** $r = -8n_t$ is a leading-order slow-roll relation; the observed several-percent violations are consistent with next-to-leading-order corrections, but this project has not computed those corrections independently, so the size of the violation is reported without a next-to-leading-order comparison to confirm it is not, instead, a numerical-tolerance artifact.

- **Arbitrary potentials are not yet systematically validated.** As before, quadratic inflation remains the only potential with a fully closed-form analytic benchmark; all others are cross-checked only against their own leading-order slow-roll formulas evaluated at the numerical background trajectory.
- **Symmetry-breaking sweep numerics.** The $\mu = 5$ point in the symmetry-breaking sweep triggered floating-point overflow warnings during root-finding (Sec. 5); the tabulated result is still self-consistent (probed against ϕ_N boundary conditions) but the reliability of the $\mu \to$ small end of that track has not been separately stress-tested.
- **Only one N_{end} benchmark tested.** All results here use $N_{\text{end}} = 65$; since $N_* = 50$ is fixed by construction, changing N_{end} affects only the trajectory between horizon crossing and the end of inflation, which is why the quadratic pivot values matched exactly between an earlier $N_{\text{end}} = 60$ pass and the current one – but this has only been explicitly checked for quadratic, not for the other seven potentials.

8 Research Experience

As with the earlier phase of this project, most of the required physics (general relativity, inflationary dynamics, cosmological perturbation theory) was not available from prior coursework and was learned specifically for this project, primarily from Baumann's notes, with regular mentor meetings providing direction and correction. A substantial part of this later phase was methodological: extending a validated single-potential notebook to eight potentials and six parameter sweeps required imposing shared conventions (a common N_{end}, a common calibration target, a common tolerance) that the original single-potential version did not need, and several early attempts at cross-potential comparison surfaced inconsistencies (such as the $N_{\text{end}} = 60$ vs. 65 mismatch between the single-potential and sweep notebooks) that had to be tracked down and resolved before the comparison was meaningful. Working through the classification hypothesis in Sec. 5 was also a lesson in distinguishing a genuine numerical pattern from an over-eager conclusion: an earlier, simpler version of the hypothesis was disproved by the non-monotonic $r(g)$ behaviour of the quadratic+quartic mixture, which is why the classification is presented here explicitly as preliminary rather than as a finished result.

9 Conclusion

This project extended a validated single-field numerical inflation pipeline from one potential (quadratic, scalar-only) to eight potentials with both scalar and tensor perturbations, all calibrated to a shared $N_{\text{end}} = 65$, $\mathcal{P}_{\mathcal{R}}(k_*) = 2.05 \times 10^{-9}$ benchmark. The scalar sector remains validated against closed-form quadratic slow-roll expectations to sub-percent accuracy; the tensor sector and the single-field consistency relation $r = -8n_t$ were checked across all eight potentials and show violations that track a potential's $|\eta_v/\varepsilon_v|$ at the pivot, largest for the hilltop-type potentials. Systematic parameter sweeps reveal a classification of a potential's free parameters into scale-type and shape-type, with distinct signatures in the (n_s, r) plane, observed and numerically verified across all seven sweeps –

this is the leading candidate for the project's next phase, with an analytic derivation of the pattern as the natural next step.

References

[1] D. Baumann, *Cosmology*, lecture notes. Primary theoretical reference throughout, particularly the sections on inflation, cosmological perturbation theory, and quantum initial conditions.